Quantum sampling for a long/short stock split: an honest benchmark of QAOA variants on 12–29 qubits

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Abstract

We encode the choice of a low-risk long/short (or buy/skip) stock portfolio as an Ising problem with one qubit per asset, solve it exactly and with classical heuristics at 12 assets, and benchmark seven quantum circuit families (linear-ramp QAOA, free-angle QAOA with five optimisers, CVaR, multi-angle, counterdiabatic, two imaginary-time methods and a constraint-preserving XY mixer) on an exact statevector simulator against the same answer key. Tuned circuits sample the optimal portfolio 30–236 times more often than uniform guessing and, in long/short mode, return it as their single most likely answer; they do not beat greedy descent or simulated annealing. We report a sign error in the linear-ramp recipe as commonly written, measure the all-to-all connectivity advantage (66 vs. 285 two-qubit gates per layer at 12 qubits, 244 vs. 1 576 at 29), and show that angles tuned at 12 qubits transfer to 16 and 20. All optimised portfolios reduced out-of-sample risk; none beat the market's return. No quantum advantage is claimed; no hardware run is reported here.

1 Problem and data

Each asset i gets a spin $s_i \in \{+1, -1\}$. Mode A (long/short): $+1$ = buy, -1 = short, weights s_i/n. Mode B (buy/skip): $x_i = (1+s_i)/2 \in \{0, 1\}$, weight x_i/n, the rest in cash. With training-window mean weekly log returns μ and sample covariance C, the score to minimise is

$$J = \frac{w^{\mathsf{T}} C\, w}{V_{\mathrm{ref}}} - \eta\, \frac{\mu^{\mathsf{T}} w}{\sigma_{\mathrm{ref}}}, \qquad V_{\mathrm{ref}} = \overline{\mathrm{diag}\, C}, \tag{1}$$

with $\eta \in \{0.25, 0.5, 1, 2\}$ ($\eta = 1$ default) and an optional balance constraint (equal numbers on each side). J is an Ising energy $\sum_{i<j} J_{ij} s_i s_j + \sum_i h_i s_i + c$ exactly (verified to 10^{-16}); thinning the 29-asset couplings to the strongest 60 % by $|\rho_{ij}|$ gives a surrogate within 0.013 of the true score (rank correlation 0.975).

Data. Kaggle "Huge Stock Market Dataset" (CC0): 29 named US assets in 10 groups (tech, gold & silver, energy, banks, bonds, healthcare, staples, utilities, real estate, copper) plus SPY as a reference, weekly closes 2006-05-26 to 2017-11-10 (599 weeks). Chronological split: 418 training weeks (to 2014-05-30, containing the 2008 crash) and 180 held-out weeks. Nothing was tuned on the test weeks. The 12-asset instance is one asset per group in a fixed round-robin: AAPL, GLD, XOM, JPM, TLT, JNJ, PG, XLU, VNQ, FCX, MSFT, SLV. Bonds are the only group negatively correlated with the rest (mean $\rho \approx -0.22$; TLT–XLK -0.45); gold is near zero.

2 Methods

Answer key and classical baselines. At 12 assets all $2^{12} = 4\,096$ portfolios are enumerated (one optimum per setting). Baselines, written in NumPy: uniform random sampling, greedy single-flip/swap descent and simulated annealing, three seeds each; balance is a verified penalty $\lambda(\sum_i s_i)^2$ at 12 assets and a hard constraint at 29.

Circuit. $H^{\otimes n}$, then per layer k: $R_{zz}(2\gamma_k J_{ij}/\mathrm{scale})$ on every kept link, $R_z(2\gamma_k h_i/\mathrm{scale})$ on every asset, $R_x(2\beta_k)$ on every qubit, scale $= \max(|J|, |h|)$. Linear-ramp QAOA (LR-QAOA) [1] uses $\gamma_k = \frac{k}{p}\Delta\gamma$, $\beta_k = (1 - \frac{k-1}{p})\Delta\beta$ and was grid-tuned (15×20 points, $p = 1..5$) by lowest expected energy, never by the optimum's probability. Free-angle QAOA ($2p$ angles) was tuned by COBYLA, Nelder–Mead, Powell, L-BFGS-B and our own SPSA from the same five starts with a cap of $200p$ circuit evaluations. Extras at the same budget: CVaR objective ($\alpha = 0.1, 0.25$), multi-angle ($3p$), first-order counterdiabatic R_y layer (DC-QAOA, cf. [3]), VarQITE angle training (McLachlan), varQITE with the Pauli-term update rule of IonQ/ORNL [2] (ansatz A: R_y+CX per link; ansatz B: R_{zz} per link + R_x layer; implemented from the description, no code reused), and an XY mixer with a Dicke start for the balance constraint. An independent NumPy statevector simulator (1.7 ms per $p = 2$ circuit) agrees with Qiskit 2.5.2 `Statevector` to 3×10^{-16}. The full run (118 unit tests) took 121 s on 8 cores.

Sign of the ramp. With the mixer $R_x(+2\beta)$, lowering the energy requires γ and β of *opposite* sign. The recipe as commonly written ($\Delta\gamma = 0.6, \Delta\beta = 0.3$) drives the state towards the *highest* energies: quality 0.45 at $p = 5$ against 0.81 for uniform sampling (mode A). All ramps below have $\Delta\beta < 0$.

3 Results (12 qubits, $\eta = 1$, exact statevector)

Metrics: P_{opt} = probability that one shot is the optimal portfolio (uniform: $1/4096 = 0.00024$); quality $= (E_{\max} - \langle E\rangle)/(E_{\max} - E_{\min})$ of the sampled distribution.

Table 1: Main benchmark. Classical rows give the hit rate of the optimum per try (random: 1 000 tries; greedy and annealing: 30 restarts, three seeds).

	Mode A		Mode B	
Method	P_{opt}	quality	P_{opt}	quality
$p = 0$ control (uniform)	0.0002	0.807	0.0002	0.714
LR-QAOA, best p (5 / 4)	**0.0073**	0.950	**0.0207**	0.973
free-angle QAOA, best p (3 / 2)	0.0042	0.935	0.0122	0.976
CVaR α=0.1, $p = 3$	0.0066	0.918	**0.0577**	0.948
multi-angle, $p = 3$	0.0054	0.945	0.0057	0.987
counterdiabatic, $p = 3$	0.0067	0.938	0.0000	0.985
VarQITE (McLachlan), $p = 3$	0.0042	0.935	0.0258	0.978
varQITE Pauli-update [2]	$<10^{-5}$	0.880	$<10^{-5}$	0.871
random guessing (per try)	0.000	–	0.000	–
greedy descent (per try)	0.067	–	1.000	–
simulated annealing (per try)	0.256	–	1.000	–

Tuned circuits beat uniform sampling by 30× (A) and 85× (B); the CVaR objective reaches 236× in mode B at the price of a lower mean quality. In 1 000 shots the optimum appears at least once with probability 0.999 (A) and 1.000 (B). In mode A the circuit's single most likely portfolio *is* the optimum (rank 1 of 4 096; long AAPL, MSFT, GLD, SLV, JPM, TLT, PG, XLU; short XOM, JNJ, VNQ, FCX; Fig. 2); in mode B it is "buy TLT only",

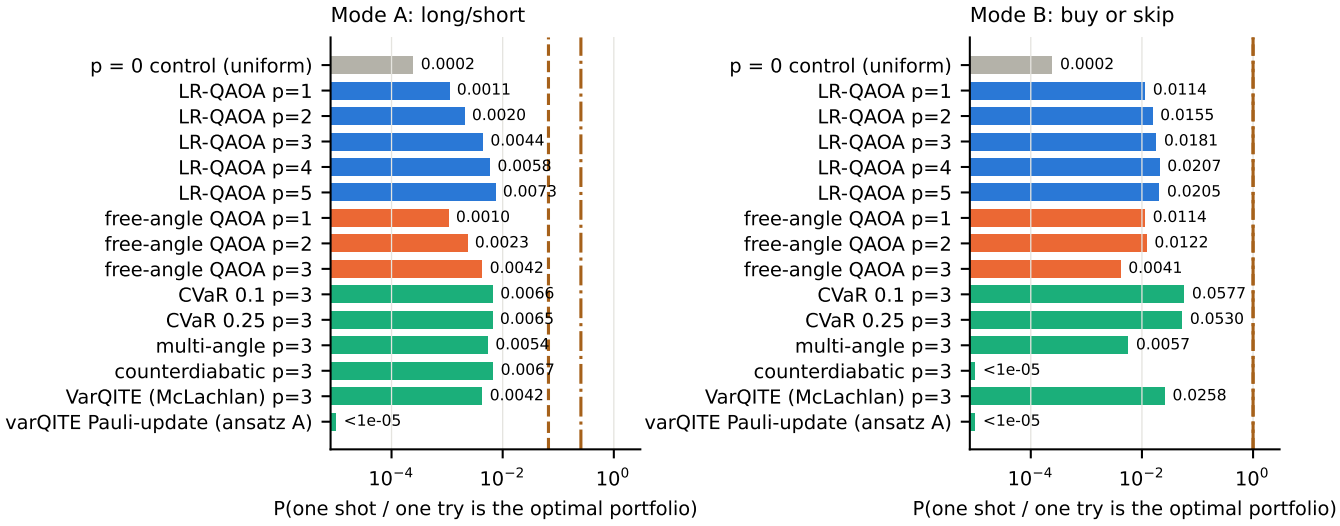


Figure 1: Every method on one scale: probability that a single shot (or classical try) is the optimal portfolio, log axis. Grey: uniform control; blue: LR-QAOA; orange: free-angle QAOA; green: the extras. Brown lines: classical hit rate per try, dotted = greedy (0.07 in A, 1.00 in B), dash-dot = simulated annealing (0.26 in A, 1.00 in B).

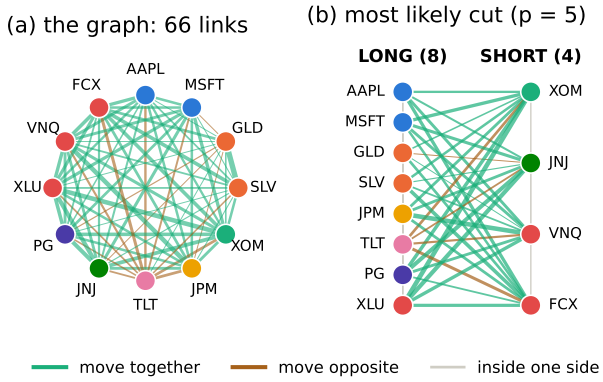


Figure 2: Mode A, 12 assets. (a) Training correlations: teal = move together, brown = move opposite, width $\propto |\rho|$. (b) The most likely LR-QAOA portfolio: 27 of 52 "together" links are cut (hedged), 5 of 14 "opposite" links.

one asset from the optimum "buy AAPL+TLT" (rank 6). No circuit approaches greedy or annealing, which hit the optimum in most tries: on this problem the circuit is a better-than-random sampler, not a better optimiser. The optimiser matters less than the start: from the tuned ramp all five reach about the same quality (mean over starts 0.86–0.97), random starts often stall; L-BFGS-B needs the fewest evaluations (24–98). The counterdiabatic layer in mode B concentrates on near-optimal states but never on the optimum. varQITE learns slowly at the published learning rates; its IonQ-native ansatz B is a stationary point at $\theta = 0$ (all parameter-shift derivatives vanish) and needs a random start. Exact imaginary time on a diagonal cost is classical Boltzmann sampling; as a reference it overtakes the best LR-QAOA at $\tau = 80$ (A) / 160 (B).

Balance. The penalty coupling 2λ is 12× (A) / 5× (B) the largest objective term, so penalised circuits mostly learn the constraint. At $p = 2$ the probability of the best *balanced* portfolio per shot is 0.0011 with the penalty (97 % balanced shots), 0.0019 (A) / 0.0009 (B) with post-selection of the unconstrained circuit (30 % / 1 % balanced), and **0.0192** (A) / **0.0068** (B) with the XY mixer (100 % balanced, 12 XX+YY gates per layer; Dicke preparation not gate-optimised).

Gate counts and transfer. One layer needs 66 two-qubit gates (depth 15) at 12 qubits and 244 (depth 29) at 29 qubits on all-to-all connectivity, where R_{zz} is native on IonQ Forte; routed on a heavy-hex grid (Qiskit transpiler, level 1, five seeds) the same layers need 285 and 1 576 (4.3× and 6.5×). The 29-qubit circuits were built and counted only. The ramp tuned at 12 assets keeps its quality at 16 (0.905 vs. 0.910 re-tuned) and 20 assets (0.919; 0.882 when thinned to 60 %), which justifies using it untuned at 29.

Out-of-sample finance (Task 3, classical optimum). Over the 180 held-out weeks, all 32 settings (2 modes × 4 η × balance on/off) had lower weekly volatility than SPY (0.00–0.91 % vs. 1.57 %) and than random portfolios of the same kind; 14 of 32 earned more than a random portfolio; none beat SPY's return (+0.165 vs. +0.214 % per week). Costs, borrow fees and slippage are not modelled; the universe has survivorship bias.

4 Limitations and outlook

Everything above is an exact classical simulation; the IonQ noise-model run (one batched job with the $p = 0$ control, LR-QAOA, CVaR and XY circuits; 12 qubits live, 29 recorded) is the next step and will decide how much of the 30–236× gain survives 66–244 two-qubit gates per layer. Deferred: warm starts, variational real-time evolution, recursive QAOA. Code, tests and all run records (including every unhelpful variant) are in the team repository.

References

[1] Linear-ramp QAOA schedule, *npj Quantum Information* (2025). [Full reference to be completed.]

[2] IonQ and ORNL, variational quantum imaginary-time evolution with a Pauli-term update rule, arXiv:2404.16135.

[3] IonQ and Kipu Quantum, portfolio optimisation on IonQ hardware, arXiv:2602.23976.